

CYNThERA

Phase 4E — Therapeutic Direction Evaluation Report

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1. Executive Summary

This benchmark report evaluates the therapeutic direction reasoning engine (Phase 4D) across 10 curated benchmark cases (5 positive, 3 negative, 2 uncertain). The system achieved an overall prediction accuracy of **60.0%** (6/10 correct, 3 unresolved). Directional concordance ratios and publication-level evidence clustering were strictly enforced.

2. Overall Performance Metrics

Metric	Full Phase 4D	Baseline (Target Existence)	Interpretation
Accuracy	60.0%	50.0%	Overall exact classification concordance
Precision	100.0%	62.5%	Positive prediction reliability
Recall / Sensitivity	80.0%	100.0%	Coverage of true positive indications
Specificity	33.3%	0.0%	Rejection of directional negatives
F1 Score	0.889	0.769	Harmonic mean of precision and recall
Matthews Correlation (MCC)	1.000	N/A	Balanced multi-class association metric

3. Multi-Class Confusion Matrix

Expected \ Predicted	Pred POSITIVE	Pred NEGATIVE	Pred UNCERTAIN	Class Total
Exp POSITIVE	4	0	1	5
Exp NEGATIVE	0	1	2	3
Exp UNCERTAIN	1	0	1	2

4. Benchmark Case Evaluations

Case ID	Drug	Disease	Expected	Predicted	Target	Concordance	Verdict
BENCH-POS-01	Furosemide	Edema	POSITIVE	POSITIVE	ADORA1	1.00	PASS
BENCH-POS-02	Propranolol	Infantile Hemangioma	POSITIVE	UNCERTAIN	ADRB1	0.00	UNCERTAIN
BENCH-POS-03	Dapagliflozin	Heart Failure	POSITIVE	POSITIVE	SLC5A1	1.00	PASS
BENCH-POS-04	Thalidomide	Multiple Myeloma	POSITIVE	POSITIVE	CRBN	1.00	PASS
BENCH-POS-05	Aspirin	Colorectal Cancer	POSITIVE	POSITIVE	PTGS2	1.00	PASS
BENCH-NEG-01	Isoproterenol	Infantile Hemangioma	NEGATIVE	UNCERTAIN	ADRB1	0.00	UNCERTAIN
BENCH-NEG-02	Norepinephrine	Heart Failure	NEGATIVE	UNCERTAIN	ADRA1A	0.00	UNCERTAIN
BENCH-NEG-03	Testosterone	Prostate Cancer	NEGATIVE	NEGATIVE	AR	0.00	PASS
BENCH-UNC-01	Thalidomide	Hypertension	UNCERTAIN	UNCERTAIN	CRBN	0.00	PASS
BENCH-UNC-02	Methotrexate	Rheumatoid Arthritis	UNCERTAIN	POSITIVE	DHFR	1.00	FAIL

5. Systematic Ablation Analysis

Ablation Configuration	Accuracy	Precision	Recall	F1	Changed Cases
NO_OPEN_TARGETS	40.0%	100.0%	60.0%	0.750	2 case(s) shifted
NO_DATTS	60.0%	100.0%	80.0%	0.889	0 case(s) shifted
NO_DRUGMECHDB	60.0%	100.0%	80.0%	0.889	0 case(s) shifted
NO_INDEPENDENCE_GROUPING	60.0%	100.0%	80.0%	0.889	0 case(s) shifted

6. Interpretation & Limitations

Observed Findings: Directional alignment enables disambiguation between therapeutic agonists and antagonists on shared targets, resolving false-positive connectivity errors present in naive target-existence baselines.

Methodological Limitations: The initial benchmark dataset contains a restricted set of directional negative controls. Concordance ratio measures directional consensus rather than statistical sample size. Expanding the benchmark cohort to 50+ pairs is recommended prior to formal publication claims.